Question 1

1.a

The accepted string in L should be simplified first. Observe that since $w = (aba)^x$ for any $x \geq 0$, we can rewrite w^R as:

$$w^R \equiv ((aba)^x)^R \equiv ((aba)^R)^x) \equiv (aba)^x \equiv w$$

And L can be modified to:

$$L = \{www | w \in (aba)^*\}$$
$$L = \{w^3 | w \in (aba)^*\}$$

Assume L is regular, then there exists a pumping length k for L. Let's choose $s = (aba)^{3k}$, $|s| \geq k$; $s \in L$ since $(aba)^{3k} \equiv ((aba)^k)^3 \in L$.

Observe all decompositions of $w = xyz$ such that $|x| \leq k$ and $|y| \geq 1$:

$$x = (aba)^n,\ n \geq 0$$
$$y = (aba)^{3m},\ m \geq 1,\ n + m \leq k$$
$$z = (aba)^{3k-3m-n}$$

Choose q such that $xy^qz \notin L$:

$$s = xy^qz = (aba)^n(aba)^{3mq}(aba)^{3k-3m-n} = (aba)^{3(k+m(q-1))}$$

Note that $s \in L$, $\forall q \geq 0$. This is because regardless of q's value, aba will always be multiplied by a multiple of 3 in the end, due to the presence of 3 on one side of the multiplication of aba's exponent, meaning aba's exponent will always be a multiple of 3. Therefore the Pumping Lemma does not work here, in fact we can derive the regular expression for L since it is regular:

$$L = (abaabaaba)^*$$

1.b

Assume L is regular, then there exists a pumping length k for L. We can choose arbitrary string $s = (01)^k(10)^k$, $|s| > k$; $s \in L$ since the number of 01 and 10 in s are both k.

Observe all decompositions of $s = xyz$ such that $|x| \leq k$ and $|y| \geq 1$:

$$x = (01)^n,\ n \geq 0$$
$$y = (01)^m,\ m \geq 1,\ n + m \leq k$$
$$z = (01)^{k-m-n}(10)^k$$

Choose q such that $xy^q z \notin L$:

$$s = (01)^n(01)^{mq}(01)^{k-m-n}(10)^k = (01)^{k+m(q-1)}(10)^k$$

If $q = 0$, then $s = (01)^{k-m}(10)^k$. $s \in L$ because in every pair of (10) there contains one (01) (i.e. $01 \in 1010$), and in every pair of (01) there exists one (10) (i.e. $10 \in 0101$).

So if we have $(k - m)$ number of (01) then inside there must exist $(k - m - 1)$ number of (10), and the count of (10) is increased to $(k+k-m-1) = 2k-m-1$.

The reverse is true for k number of (10), since we can count $(k-m-1)$ number of (10) inside the $(k-m)$ number of (01), the count of (10) is also $(k+k-m-1) = 2k - m - 1$. Then we see the number of both (10) and (01) are actually the same regardless of their exponents, and the same exists for $q = 2$.

So the pumping lemma fails. In fact, we can prove that L is regular by giving its regular expression. The regular expression is trivial to produce since we know already that $\epsilon, 0, 1$ are already $\in L$ (vacuously true) and our previously testing gives the condition that $w \in L$ if it begins and ends by the same letter:

$$L = \epsilon \cup 1 \cup 0 \cup 0(1 \cup 0)^*0 \cup 1(1 \cup 0)^*1$$